Baseline appendix

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A Full-evaluation protocol and corrections

All benchmark tables and figures in this paper use the original 28 relations, comprising 9,747 rows per model before the stated year-scoring exception. Original strings, labels, row order within the recorded selections, and duplicates are preserved. There is no correctness filter, identifying-first-token filter, row cap or minimum-test-size cohort.

Demonstrations, references and evaluation. Each method uses eight demonstration rows and eight reference rows. The LRE selection reproduces the original sequential shuffles with seeds 0 and 1000; the lens selection uses seed 17. Every original row is then evaluated, including both selections. For the lens, reference comparisons sharing the query's subject string are omitted from its average; the query is still scored, and duplicates remain in aggregate statistics. LRE retains its original reference estimator. The different demonstration contexts can produce different target tokens, which is recorded rather than silently treated as a matched-context evaluation.

LRE selection. The primary outcome chooses a layer and scaling from the original six-by-six grid on the full evaluation. Table 3 also reports the original fixed layer/scaling evaluated on every row. Neither version is an independent held-out estimate. Full-data LRE measurements and target tokens are reused for neutral-prompt changes; only affected lens readouts are recomputed.

Gender wording. For a target in the female wording set, the accepted single-token, leading-space terms are `woman, her, herself, she, hers, female, girl`; for the male set they are `man, him, himself, he, his, male, boy`. A lens rank is the best rank in the matching set; other model targets remain unchanged. This rule is applied to every vocabulary-rank readout in the baseline comparison. Dataset labels are not rewritten. LRE faithfulness still requires its exact top-1 token to match the model's top-1; the alternative wordings modify the lens axis.

Year tokenization. The answer boundary is checked with each actual tokenizer and the joint prompt–answer string. Where necessary, a trailing space is consumed in the relation prompt so that the next token is content. A standalone first answer token `1` is distinguished from multi-digit tokens such as `178`. Under the relation-level rule used for the primary comparison, PRESIDENTS_BIRTH_YEAR and PRESIDENTS_ELECTION_YEAR are excluded for Mistral, Gemma and both Qwen models, leaving 26 scored relations and 9,709 rows each. The artifact occurs on all 19 birth-year rows and 16 of the 19 election-year rows in each affected model. The relation-level rule removes both complete year relations, including the other three election-year rows, from the primary correlation. The rule does not depend on measured faithfulness. OLMo, GPT-J and Pythia retain both year relations and all 9,747 rows. All original rows and measurements remain available; the all-28 column in Table 3 shows the year-inclusive sensitivity.

Continent references. We retain the original seeded references for LANDMARK_ON_CONTINENT; there is no diversity correction in these reported values. Their selection and category counts are stored with each measurement. This setting is applied consistently to both methods and all models.

Table 1: Full-evaluation contrastive/differential Jacobian lens correlations with LRE faithfulness. LRE layer and scaling are selected on the full evaluation grid.

Model	Relations	Scored rows	Spearman ρ	p
OLMo 3 7B	28	9,747	0.7355	8.24×10^{-6}
GPT-J 6B†	28	9,747	0.4346	0.021
Pythia 6.9B†	28	9,747	0.5027	0.006
Mistral 7B	26	9,709	0.6561	2.74×10^{-4}
Gemma 2 9B	26	9,709	0.6889	9.96×10^{-5}
Qwen3 8B Base†	26	9,709	0.6647	2.12×10^{-4}
Qwen3.8 27B†	26	9,709	0.5539	0.003

†Leading newline in the neutral lens prompt to address high first-position activation norms. All eligible rows, including demonstration/reference rows, are evaluated. Models with 26 scored relations exclude the two year relations whose first answer token is the standalone digit 1; all 28 relations were measured. Gender alternatives are enabled; continent references use the original seeded selection.

Table 2: Full-evaluation newline ablation for the two additional models. All 28 relations, 9,747 rows, LRE results and target tokens are fixed.

Model	Original ρ	Newline ρ	Original norm	Newline norm
GPT-J 6B	0.1883	0.4346	4209	123
Pythia 6.9B	0.1746	0.5027	2876	282

Norm columns give median subject-state norms for rows whose subject endpoint was at token position zero before adding the newline (142 GPT-J rows; 83 Pythia rows). The newline shifts the subject away from position zero without changing its content token IDs. Qwen3 8B Base and Qwen3.8 27B retain the previously adopted newline correction; their current full-evaluation correlations are in Table 1.

Leading newline and high norms. GPT-J 6B, Pythia 6.9B, Qwen3 8B Base and Qwen3.8 27B use `\n{} is known for`. A dagger marks these models in the result tables. OLMo, Mistral and Gemma use `{} is known for`. The newline addresses unusually large subject-state norms when content occupies the first token position. Table 2 reports the additional full-row GPT-J and Pythia ablations. Qwen's previously adopted correction is retained. The ablations change neither the LRE outcome nor its target tokens.

Numerical checks and provenance. Measurements use pinned checkpoint revisions, `float32`, eager attention, and the architecture's actual final normalization and output head. Scalar and batched Jacobian–vector products, finite differences, and cached computations are numerically checked. The report validates complete row multisets, duplicate counts, target IDs, reference counts, and source hashes, and recomputes aggregates from per-row ranks. Current numbers are generated from these records rather than copied from historical tables. Earlier shared-split, capped, correctness-filtered, and separate small-cohort numerical analyses are archived outside the compiled paper and are not used to substantiate its current claims.

B Baseline comparisons

Readout definitions. The contrastive/differential Jacobian lens (CEL) decodes $\mathrm{mean}_i\, J_i(s_e - s_i)$; the noncontrastive/raw Jacobian lens (NEL) decodes $\mathrm{mean}_i\, J_i s_e$. Logit lens decodes s_e, and differential logit lens decodes $s_e - \mathrm{mean}_i\, s_i$. Every readout applies the same final normalization and output head. Scores are the negative median of $\log_{10}(\mathrm{rank} + 1)$ for zero-based ranks, with the gender rule above. Every eligible relation uses every original row.

Attribute rate. Following Geva et al. (2023), we compare filtered top-50 logit-lens predictions with attributes retrieved from Wikipedia. The retrieval queries cover every original subject, including those used as demonstrations or references. The source length, stopword and retrieval filters are retained. If no prediction survives the filter, the original overlap fraction is undefined; for this full-row comparison we explicitly extend it to zero and retain the row. The accompanying measurements record every such case. This convention changes the metric for those cases and avoids a model-specific filtered cohort.

Table 3: Correlation sensitivity to LRE parameter selection and year scoring. All columns use complete relation evaluations.

Model	Primary n	Selected	Frozen	All 28
OLMo 3 7B	28	0.7355	0.7046	0.7355
GPT-J 6B†	28	0.4346	0.4609	0.4346
Pythia 6.9B†	28	0.5027	0.4026	0.5027
Mistral 7B	26	0.6561	0.5986	0.3687
Gemma 2 9B	26	0.6889	0.6862	0.4541
Qwen3 8B Base†	26	0.6647	0.6490	0.4147
Qwen3.8 27B†	26	0.5539	0.5296	0.3037

Selected: best layer/scaling on full evaluation. Frozen: original layer/scaling retained, outcomes evaluated on all rows. All 28: both year relations included as a tokenization sensitivity, using selected LRE parameters. †Newline models as in Table 1.

Table 4: Neural and Wikipedia attribute baselines using the same full evaluations, targets, reference sets, layer, prompt correction, gender alternatives and year rule as the contrastive lens.

Model	CEL	NEL	Logit	Diff. logit	Attribute
OLMo 3 7B	0.7355	0.7650	0.2056	0.4390	-0.4016
GPT-J 6B†	0.4346	0.4196	0.4669	0.2020	-0.1035
Pythia 6.9B†	0.5027	0.5063	-0.2047	0.1845	0.0041
Mistral 7B	0.6561	0.6711	0.6581	0.5186	-0.2202
Gemma 2 9B	0.6889	0.6868	0.4728	0.3436	-0.5515
Qwen3 8B Base†	0.6647	0.6232	0.7908	0.5952	-0.0557
Qwen3.8 27B†	0.5539	0.5717	0.3404	0.4492	-0.0099

CEL: contrastive/differential Jacobian lens; NEL: raw/noncontrastive Jacobian lens. Higher scores mean better target ranks. †Leading newline. Attribute overlap is zero when no top-50 predictions survive the published length/stopword filter; no evaluation rows are dropped. These are point estimates, not evidence of significant differences between readouts.

B.1 OLMo corpus-frequency baselines

A document contributes at most one mention per subject/object string. Matching uses normalized literal names with Unicode word boundaries, without alias resolution or a proximity constraint. The saved subject, object and joint counts cover every original fact string. We restore original duplicate multiplicities and recompute relation aggregates before correlating them with the current LRE outcomes. Unobserved conditioning marginals receive the explicit zero convention noted in the table, and no row is dropped.

These associations remain exploratory: relations can share subjects and objects, the counts do not resolve aliases, and p-values are unadjusted. They are not evidence that training frequency causes the observed readout or LRE performance.

References

Mor Geva, Jasmijn Bastings, Katja Filippova, and Amir Globerson. Dissecting recall of factual associations in auto-regressive language models. In Houda Bouamor, Juan Pino, and Kalika Bali, editors, *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12216–12235, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.751. URL https://aclanthology.org/2023.emnlp-main.751/.

Table 5: Attribute-rate coverage under the full primary evaluation.

Model	Scored rows	Empty filtered predictions
OLMo 3 7B	9,747	0
GPT-J 6B†	9,747	0
Pythia 6.9B†	9,747	2,663
Mistral 7B	9,709	0
Gemma 2 9B	9,709	1
Qwen3 8B Base†	9,709	0
Qwen3.8 27B†	9,709	0

Rows with no surviving top-50 prediction are assigned zero overlap in the reported extension. No original row is removed for an undefined attribute rate.

Table 6: Anchored neutral Jacobian readout at fixed $\beta = 1$, under the same full evaluation and scoring corrections.

Model	Spearman ρ
OLMo 3 7B	0.4070
GPT-J 6B†	0.4548
Pythia 6.9B†	0.2282
Mistral 7B	0.3197
Gemma 2 9B	0.4044
Qwen3 8B Base†	0.4229
Qwen3.8 27B†	0.4537

The readout averages $f(s_i) + J_i(s - s_i)$ under the neutral prompt. The LRE outcome still uses its selected relation-context layer and scaling. No cross-relation tuning of the neutral readout is performed.

Table 7: OLMo corpus-frequency baselines, reaggregated over all 9,747 original rows and correlated with full-evaluation LRE faithfulness across all 28 relations.

Feature	Spearman ρ	p
Mean subject document count	0.1380	0.484
Mean object document count	0.5712	0.001
Mean joint document count	0.4568	0.015
Mean $\log(1 + c_{so})$	0.4191	0.026
$\log(1 + \text{mean}(c_{so}))$	0.4568	0.015
Mean $P(o \mid s)$	0.4486	0.017
Mean $P(s \mid o)$	−0.4149	0.028
Mean $P(o \mid s) - P(s \mid o)$	0.4826	0.009
Micro $P(o \mid s)$	0.5504	0.002
Micro $P(s \mid o)$	−0.1453	0.461

Counts use binary document presence across the three saved OLMo training-mix scans. Duplicate dataset rows receive their original multiplicity. An unobserved conditioning marginal gives a conditional feature of zero, rather than dropping the row; 207 rows have at least one unobserved marginal. Micro conditionals divide summed joint counts by summed marginal counts. Values are exploratory and p-values are unadjusted.